

Creation Date 17-Nov-2009

Revision Date 30-Nov-2024

Revision Number 4

Section 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

1.1. Product identifier

Product Description: Triethyloxonium tetrafluoroborate, 1.0M in dichloromethane
Cat No. : **A11678**
Molecular Formula C₆ H₁₅ O . B F₄

1.2. Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.
Uses advised against No Information available

1.3. Details of the supplier of the safety data sheet

Company

Thermo Fisher (Kandel) GmbH
Erlenbachweg 2, 76870 Kandel, Germany
Tel: +49 (0) 721 84007 280
Fax: +49 (0) 721 84007 300

Swiss distributor - Fisher Scientific AG
Neuhofstrasse 11, CH 4153 Reinach
Tel: +41 (0) 56 618 41 11
<https://www.fishersci.ch/ch/en/customer-help-support/forms/email-us.html>

E-mail address begel.sdsdesk@thermofisher.com

1.4. Emergency telephone number

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99
CHEMTREC Tel. No.**US**:001-800-424-9300 / **Europe**:001-703-527-3887

customers in Switzerland:
Tox Info Suisse Emergency Number: **145 (24hr)**
Tox Info Suisse: +41-44 251 51 51 (Emergency number from abroad)
Chemtrec (24h) Toll-Free: 0800 564 402
Chemtrec Local: +41-43 508 20 11 (Zurich)

Poison Centre - Emergency information services

Ireland : National Poisons Information Centre (NPIC) -
01 809 2166 (8am-10pm, 7 days a week)
Malta : +356 2395 2000
Cyprus : +357 2240 5611

Section 2: HAZARDS IDENTIFICATION

2.1. Classification of the substance or mixture

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CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

Based on available data, the classification criteria are not met

Health hazards

Skin Corrosion/Irritation

Serious Eye Damage/Eye Irritation

Carcinogenicity

Specific target organ toxicity - (single exposure)

Category 1 B (H314)

Category 1 (H318)

Category 2 (H351)

Category 3 (H336)

Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

2.2. Label elements



Signal Word

Danger

Hazard Statements

H314 - Causes severe skin burns and eye damage

H336 - May cause drowsiness or dizziness

H351 - Suspected of causing cancer

EUH014 - Reacts violently with water

Precautionary Statements

P310 - Immediately call a POISON CENTER or doctor/physician

P301 + P330 + P331 - IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

P280 - Wear protective gloves/protective clothing/eye protection/face protection

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

2.3. Other hazards

Reacts violently with water

Toxic to terrestrial vertebrates

Contains a known or suspected endocrine disruptor

Contains a substance on the National Authorities Endocrine Disruptor Lists

Section 3: Composition/information on ingredients

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3.2. Mixtures

Component	CAS No	EC No	Weight %	CLP Classification - Regulation (EC) No 1272/2008
Dichloromethane	75-09-2	EEC No. 200-838-9	80 - 90	Skin Irrit. 2 (H315) Eye Irrit. 2 (H319) STOT SE 3 (H336) Carc. 2 (H351)
Oxonium, triethyl-, tetrafluoroborate(1-)	368-39-8	EEC No. 206-705-1	10 - 20	Skin Corr. 1B (H314) Eye Dam. 1 (H318) (EUH014)

Components	Reach Registration Number
Dichloromethane	01-2119480404-41

Full text of Hazard Statements: see section 16

Section 4: First aid measures

4.1. Description of first aid measures

General Advice	Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.
Skin Contact	Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.
Ingestion	Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.
Inhalation	If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.
Self-Protection of the First Aider	Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

4.2. Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. Difficulty in breathing. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

4.3. Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically. Symptoms may be delayed.
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Section 5: Firefighting measures

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5.1. Extinguishing media

Suitable Extinguishing Media

CO₂, dry chemical, dry sand, alcohol-resistant foam.

Extinguishing media which must not be used for safety reasons

Water.

5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Reacts violently with water.

Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO₂), Oxides of boron, Hydrogen fluoride.

5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

Section 6: ACCIDENTAL RELEASE MEASURES

6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

6.2. Environmental precautions

Should not be released into the environment.

6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water.

6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

Section 7: Handling and storage

7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water.

Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

7.2. Conditions for safe storage, including any incompatibilities

Keep away from water or moist air. Corrosives area. Store in freezer. Keep under nitrogen. Keep containers tightly closed in a dry, cool and well-ventilated place.

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Technical Rules for Hazardous Substances (TRGS) 510
Storage Class (LGK) (Germany)

Storage Class/LGK 8A

Switzerland - Storage of hazardous substances

Storage class - SC 8
<https://www.kvu.ch/de/themen/stoffe-und-produkte>
<https://www.kvu.ch/fr/themes/substances-et-produits>
<https://www.kvu.ch/it/temi/sostanze-e-prodotti>

7.3. Specific end use(s)

Use in laboratories

Section 8: Exposure controls/personal protection

8.1. Control parameters

Exposure limits

List source(s): **UK** - EH40/2005 Work Exposure Limits, Forth edition. Published 2020. **IRE** - 2010 Code of Practice for the Safety, Health and Welfare at Work (Chemical Agents) Regulations 2001. Published by the Health and Safety Authority. **CH** - The Government of Switzerland has set a directive on limit values for working materials (Grenzwerte am Arbeitsplatz) which is based on the Swiss Federal Regulation "Verordnung über die Verhütung von Unfällen und Berufskrankheiten". This directive is administered, periodically revised and enforced by SUVA (Swiss National Accident Insurance Fund). **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC

Component	European Union	The United Kingdom	France	Belgium	Spain
Dichloromethane	TWA: 353 mg/m ³ (8h) TWA: 100 ppm (8h) STEL: 706 mg/m ³ (15min) STEL: 200 ppm (15min) Skin	STEL: 200 ppm 15 min STEL: 706 mg/m ³ 15 min TWA: 353 mg/m ³ 8 hr TWA: 100 ppm 8 hr Skin	TWA / VME: 50 ppm (8 heures). restrictive limit TWA / VME: 178 mg/m ³ (8 heures). restrictive limit STEL / VLCT: 100 ppm. restrictive limit STEL / VLCT: 356 mg/m ³ . restrictive limit Peau	TWA: 50 ppm 8 uren TWA: 177 mg/m ³ 8 uren STEL: 200 ppm 15 minuten STEL: 706 mg/m ³ 15 minuten Huid	STEL / VLA-EC: 100 ppm (15 minutos). STEL / VLA-EC: 353 mg/m ³ (15 minutos). TWA / VLA-ED: 50 ppm (8 horas) TWA / VLA-ED: 177 mg/m ³ (8 horas)

Component	Italy	Germany	Portugal	The Netherlands	Finland
Dichloromethane	TWA: 175 mg/m ³ 8 ore. Time Weighted Average TWA: 50 ppm 8 ore. Time Weighted Average STEL: 353 mg/m ³ 15 minuti. Short-term STEL: 100 ppm 15 minuti. Short-term Pelle	TWA: 50 ppm (8 Stunden). AGW - exposure factor 2 TWA: 180 mg/m ³ (8 Stunden). AGW - exposure factor 2 TWA: 50 ppm (8 Stunden). MAK TWA: 180 mg/m ³ (8 Stunden). MAK Höhepunkt: 100 ppm Höhepunkt: 360 mg/m ³ Haut	STEL: 706 mg/m ³ 15 minutos STEL: 200 ppm 15 minutos TWA: 353 mg/m ³ 8 horas TWA: 100 ppm 8 horas Pele	huid STEL: 200 ppm 15 minuten STEL: 706 mg/m ³ 15 minuten TWA: 100 ppm 8 uren TWA: 353 mg/m ³ 8 uren	TWA: 50 ppm 8 tunteina TWA: 177 mg/m ³ 8 tunteina STEL: 100 ppm 15 minuutteina STEL: 353 mg/m ³ 15 minuutteina Iho
Oxonium, triethyl-, tetrafluoroborate(1-)		TWA: 1 mg/m ³ (8 Stunden). AGW - exposure factor 4 TWA: 1 mg/m ³ (8 Stunden). MAK Haut	TWA: 2.5 mg/m ³ 8 horas		

Component	Austria	Denmark	Switzerland	Poland	Norway
Dichloromethane	Haut MAK-KZGW: 200 ppm 15 Minuten MAK-KZGW: 700 mg/m ³ 15 Minuten	TWA: 35 ppm 8 timer TWA: 122 mg/m ³ 8 timer STEL: 706 mg/m ³ 15 minutter STEL: 200 ppm 15	Haut/Peau STEL: 200 ppm 15 Minuten STEL: 706 mg/m ³ 15 Minuten	STEL: 353 mg/m ³ 15 minutach TWA: 88 mg/m ³ 8 godzinach	TWA: 15 ppm 8 timer TWA: 50 mg/m ³ 8 timer STEL: 45 ppm 15 minutter. value from the regulation

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	MAK-TMW: 50 ppm 8 Stunden MAK-TMW: 175 mg/m ³ 8 Stunden	minutter Hud	TWA: 50 ppm 8 Stunden TWA: 177 mg/m ³ 8 Stunden		STEL: 150 mg/m ³ 15 minutter. value from the regulation Hud
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Component	Bulgaria	Croatia	Ireland	Cyprus	Czech Republic
Dichloromethane	TWA: 353 mg/m ³ TWA: 100 ppm STEL : 706 mg/m ³ STEL : 200 ppm Skin notation	kože TWA-GVI: 100 ppm 8 satima. TWA-GVI: 353 mg/m ³ 8 satima. STEL-KGVI: 200 ppm 15 minutama. STEL-KGVI: 706 mg/m ³ 15 minutama.	TWA: 100 ppm 8 hr. TWA: 353 mg/m ³ 8 hr. STEL: 200 ppm 15 min STEL: 706 mg/m ³ 15 min Skin	Skin-potential for cutaneous absorption STEL: 706 mg/m ³ STEL: 200 ppm TWA: 353 mg/m ³ TWA: 100 ppm	TWA: 200 mg/m ³ 8 hodinách. Potential for cutaneous absorption Ceiling: 500 mg/m ³

Component	Estonia	Gibraltar	Greece	Hungary	Iceland
Dichloromethane	Nahk TWA: 35 ppm 8 tundides. TWA: 120 mg/m ³ 8 tundides. STEL: 70 ppm 15 minutites. STEL: 250 mg/m ³ 15 minutites.	Skin notation TWA: 353 mg/m ³ 8 hr TWA: 100 ppm 8 hr STEL: 706 mg/m ³ 15 min STEL: 200 ppm 15 min	skin - potential for cutaneous absorption STEL: 200 ppm STEL: 706 mg/m ³ TWA: 100 ppm TWA: 353 mg/m ³	STEL: 200 ppm 15 percekben. CK STEL: 706 mg/m ³ 15 percekben. CK TWA: 100 ppm 8 órában. AK TWA: 353 mg/m ³ 8 órában. AK lehetséges borón keresztüli felszívódás	TWA: 35 ppm 8 klukkustundum. TWA: 122 mg/m ³ 8 klukkustundum. Skin notation Ceiling: 70 ppm Ceiling: 244 mg/m ³

Component	Latvia	Lithuania	Luxembourg	Malta	Romania
Dichloromethane	skin - potential for cutaneous exposure STEL: 150 mg/m ³ STEL: 42 ppm TWA: 120 mg/m ³ TWA: 34 ppm	TWA: 35 ppm IPRD TWA: 120 mg/m ³ IPRD Oda STEL: 70 ppm STEL: 250 mg/m ³	Possibility of significant uptake through the skin TWA: 100 ppm 8 Stunden TWA: 353 mg/m ³ 8 Stunden STEL: 200 ppm 15 Minuten STEL: 706 mg/m ³ 15 Minuten	possibility of significant uptake through the skin TWA: 100 ppm TWA: 353 mg/m ³ STEL: 200 ppm 15 minuti STEL: 706 mg/m ³ 15 minuti	Skin notation TWA: 100 ppm 8 ore TWA: 353 mg/m ³ 8 ore STEL: 200 ppm 15 minute STEL: 706 mg/m ³ 15 minute

Component	Russia	Slovak Republic	Slovenia	Sweden	Turkey
Dichloromethane	TWA: 50 mg/m ³ 0922 MAC: 100 mg/m ³	Ceiling: 706 mg/m ³ Potential for cutaneous absorption TWA: 100 ppm TWA: 353 mg/m ³	TWA: 100 ppm 8 urah TWA: 353 mg/m ³ 8 urah Koža STEL: 200 ppm 15 minutah STEL: 706 mg/m ³ 15 minutah	Binding STEL: 70 ppm 15 minuter Binding STEL: 250 mg/m ³ 15 minuter TLV: 35 ppm 8 timmar. NGV TLV: 120 mg/m ³ 8 timmar. NGV Hud	

Biological limit values

List source(s): **UK** - Biological Monitoring Guidance Values provided by the UK's Health and Safety Executive (HSE) Control of Substances Hazardous to Health Regulations (COSHH) 2002 (as amended) and EH40/2005.

Component	European Union	United Kingdom	France	Spain	Germany
Dichloromethane		Carbon monoxide: 30 ppm end-tidal breath post shift	Dichloromethane: 0.2 mg/L urine end of shift Carboxyhémoglobine sanguine: 3.5 % blood end of shift	Dichloromethane: 0.3 mg/L urine end of shift	Dichloromethane: 500 µg/L whole blood (immediately after exposure)

Component	Italy	Finland	Denmark	Bulgaria	Romania
Dichloromethane					Carboxyhémoglobine: 5 % Hemoglobin blood end of shift Methylene chloride: 0.3

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					mg/L urine end of shift Methylene chloride: 1 mg/L blood end of shift
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Component	Gibraltar	Latvia	Slovak Republic	Luxembourg	Turkey
Dichloromethane			Dichloromethane: 1 mg/L blood end of exposure or work shift Carboxyhemoglobin: 5 % of hemoglobin blood end of exposure or work shift		

Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

Component	Acute effects local (Dermal)	Acute effects systemic (Dermal)	Chronic effects local (Dermal)	Chronic effects systemic (Dermal)
Dichloromethane 75-09-2 (80 - 90)				DNEL = 12mg/kg bw/day

Component	Acute effects local (Inhalation)	Acute effects systemic (Inhalation)	Chronic effects local (Inhalation)	Chronic effects systemic (Inhalation)
Dichloromethane 75-09-2 (80 - 90)		DMEL = 132.14mg/m ³		DNEL = 176mg/m ³

Predicted No Effect Concentration (PNEC)

See values below.

Component	Fresh water	Fresh water sediment	Water Intermittent	Microorganisms in sewage treatment	Soil (Agriculture)
Dichloromethane 75-09-2 (80 - 90)	PNEC = 130µg/L PNEC = 0.31mg/L	PNEC = 163µg/kg sediment dw PNEC = 2.57mg/kg sediment dw	PNEC = 0.27mg/L	PNEC = 26mg/L	PNEC = 173µg/kg soil dw PNEC = 0.33mg/kg soil dw

Component	Marine water	Marine water sediment	Marine water Intermittent	Food chain	Air
Dichloromethane 75-09-2 (80 - 90)	PNEC = 130µg/L PNEC = 0.031mg/L	PNEC = 163µg/kg sediment dw PNEC = 0.26mg/kg sediment dw	PNEC = 0.027mg/L		

8.2. Exposure controls

Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

Personal protective equipment

Eye Protection

Goggles (European standard - EN 166)

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Hand Protection

Protective gloves

Glove material	Breakthrough time	Glove thickness	EU standard	Glove comments
Viton (R)	See manufacturers recommendations	-	EN 374	(minimum requirement)

Skin and body protection

Long sleeved clothing.

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced
Recommended Filter type: low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387

Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.
Recommended half mask:- Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141
When RPE is used a face piece Fit Test should be conducted

Environmental exposure controls

No information available.

Section 9: Physical and chemical properties

9.1. Information on basic physical and chemical properties

Physical State	Liquid	
Appearance	Light yellow	
Odor	No information available	
Odor Threshold	No data available	
Melting Point/Range	No data available	
Softening Point	No data available	
Boiling Point/Range	No information available	
Flammability (liquid)	No data available	
Flammability (solid,gas)	Not applicable	Liquid
Explosion Limits	No data available	
Flash Point	No information available	Method - No information available
Autoignition Temperature	No data available	
Decomposition Temperature	No data available	
pH	No information available	
Viscosity	No data available	
Water Solubility	Reacts violently with water	
Solubility in other solvents	No information available	
Partition Coefficient (n-octanol/water)		
Component	log Pow	
Dichloromethane	1.25	
Vapor Pressure	No information available	
Density / Specific Gravity	1.328	
Bulk Density	Not applicable	Liquid

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Vapor Density No information available (Air = 1.0)
Particle characteristics (liquid) Not applicable

9.2. Other information

Molecular Formula C6 H15 O . B F4
Molecular Weight 189.99
Evaporation Rate No information available

Section 10: Stability and reactivity

10.1. Reactivity Yes

10.2. Chemical stability Reacts violently with water.

10.3. Possibility of hazardous reactions

Hazardous Polymerization Hazardous polymerization does not occur.
Hazardous Reactions None under normal processing. Reacts violently with water.

10.4. Conditions to avoid Incompatible products. Excess heat. Exposure to moist air or water. Exposure to moisture.

10.5. Incompatible materials Strong oxidizing agents. Bases. Strong acids. Amines.

10.6. Hazardous decomposition products Carbon monoxide (CO). Carbon dioxide (CO₂). Oxides of boron. Hydrogen fluoride.

Section 11: Toxicological information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Product Information

(a) acute toxicity;
Oral Based on available data, the classification criteria are not met
Dermal Based on available data, the classification criteria are not met
Inhalation Based on available data, the classification criteria are not met

Toxicology data for the components

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation
Dichloromethane	> 2000 mg/kg (Rat)	> 2000 mg/kg (Rat)	53 mg/L (Rat) 6 h 76000 mg/m ³ (Rat) 4 h

(b) skin corrosion/irritation; Category 1 B

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;
Respiratory No data available

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Skin	No data available										
(e) germ cell mutagenicity;	No data available										
(f) carcinogenicity;	Category 2 The table below indicates whether each agency has listed any ingredient as a carcinogen										
<table><tr><th>Component</th><th>EU</th><th>UK</th><th>Germany</th><th>IARC</th></tr><tr><td>Dichloromethane</td><td></td><td></td><td></td><td>Group 2A</td></tr></table>		Component	EU	UK	Germany	IARC	Dichloromethane				Group 2A
Component	EU	UK	Germany	IARC							
Dichloromethane				Group 2A							
(g) reproductive toxicity;	No data available										
(h) STOT-single exposure;	Category 3										
Results / Target organs	Central nervous system (CNS).										
(i) STOT-repeated exposure;	No data available										
Target Organs	No information available.										
(j) aspiration hazard;	No data available										
Symptoms / effects,both acute and delayed	Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.										

11.2. Information on other hazards

Endocrine Disrupting Properties
Assess endocrine disrupting properties for human health

Contains a substance on the National Authorities Endocrine Disruptor Lists

Section 12: Ecological information

12.1. Toxicity

Ecotoxicity effects

Do not empty into drains. Reacts with water so no ecotoxicity data for the substance is available.

Component	Freshwater Fish	Water Flea	Freshwater Algae
Dichloromethane	Pimephales promelas: LC50:193 mg/L/96h	EC50: 140 mg/L/48h	EC50:>660 mg/L/96h

Component	Microtox	M-Factor
Dichloromethane	EC50: 1 mg/L/24 h EC50: 2.88 mg/L/15 min	

12.2. Persistence and degradability

Persistence
Degradability
Degradation in sewage treatment plant

No information available
Persistence is unlikely, based on information available.
Reacts with water.
Reacts violently with water.

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12.3. Bioaccumulative potential Product does not bioaccumulate due to reaction with water; Bioaccumulation is unlikely

Component	log Pow	Bioconcentration factor (BCF)
Dichloromethane	1.25	6.4 - 40 dimensionless

12.4. Mobility in soil Reacts violently with water . Is not likely mobile in the environment.

12.5. Results of PBT and vPvB assessment Reacts violently with water.

12.6. Endocrine disrupting properties
Endocrine Disruptor Information This product does not contain any known or suspected endocrine disruptors

12.7. Other adverse effects
Persistent Organic Pollutant This product does not contain any known or suspected substance
Ozone Depletion Potential This product does not contain any known or suspected substance

Section 13: Disposal considerations

13.1. Waste treatment methods

Waste from Residues/Unused Products Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

Contaminated Packaging Dispose of this container to hazardous or special waste collection point.

European Waste Catalogue (EWC) According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

Other Information Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not flush to sewer. Large amounts will affect pH and harm aquatic organisms.

Switzerland - Waste Ordinance Disposal should be in accordance with applicable regional, national and local laws and regulations. Ordinance on the Avoidance and the Disposal of Waste (Waste Ordinance, ADWO) SR 814.600
<https://www.fedlex.admin.ch/eli/cc/2015/891/en>

Section 14: Transport information

IMDG/IMO

14.1. UN number UN2922
14.2. UN proper shipping name Corrosive liquid, toxic, n.o.s.
Technical Shipping Name (TRIETHYLOXONIUM TETRAFLUOROBORATE, METHYLENE CHLORIDE)
14.3. Transport hazard class(es) 8
Subsidiary Hazard Class 6.1
14.4. Packing group II

ADR

14.1. UN number UN2922
14.2. UN proper shipping name Corrosive liquid, toxic, n.o.s.
Technical Shipping Name (TRIETHYLOXONIUM TETRAFLUOROBORATE, METHYLENE CHLORIDE)

ALFAAA11678

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14.3. Transport hazard class(es) 8
Subsidiary Hazard Class 6.1
14.4. Packing group II

IATA

14.1. UN number UN2922
14.2. UN proper shipping name Corrosive liquid, toxic, n.o.s.
Technical Shipping Name (TRIETHYLOXONIUM TETRAFLUOROBORATE, METHYLENE CHLORIDE)
14.3. Transport hazard class(es) 8
Subsidiary Hazard Class 6.1
14.4. Packing group II

14.5. Environmental hazards No hazards identified

14.6. Special precautions for user No special precautions required.

14.7. Maritime transport in bulk according to IMO instruments Not applicable, packaged goods

Section 15: Regulatory information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

Component	CAS No	EINECS	ELINCS	NLP	IECSC	TCSI	KECL	ENCS	ISHL
Dichloromethane	75-09-2	200-838-9	-	-	X	X	KE-23893	X	X
Oxonium, triethyl-, tetrafluoroborate(1-)	368-39-8	206-705-1	-	-	-	X	-	-	X

Component	CAS No	TSCA	TSCA Inventory notification - Active-Inactive	DSL	NDSL	AICS	NZIoC	PICCS
Dichloromethane	75-09-2	X	ACTIVE	X	-	X	X	X
Oxonium, triethyl-, tetrafluoroborate(1-)	368-39-8	X	ACTIVE	X	-	-	X	-

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

Authorisation/Restrictions according to EU REACH

Component	CAS No	REACH (1907/2006) - Annex XIV - Substances Subject to Authorization	REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances	REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC)
Dichloromethane	75-09-2	-	Use restricted. See entry 59. (see link for restriction details) Use restricted. See entry 75. (see link for restriction details)	-
Oxonium, triethyl-, tetrafluoroborate(1-)	368-39-8	-	-	-

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REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

Seveso III Directive (2012/18/EC)

Component	CAS No	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification	Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements
Dichloromethane	75-09-2	Not applicable	Not applicable
Oxonium, triethyl-, tetrafluoroborate(1-)	368-39-8	Not applicable	Not applicable

Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

WGK Classification

Water endangering class = 2 (self classification)

Component	Germany - Water Classification (AwSV)	Germany - TA-Luft Class
Dichloromethane	WGK2	Class I : 20 mg/m ³ (Massenkonzentration)

Component	France - INRS (Tables of occupational diseases)
Dichloromethane	Tableaux des maladies professionnelles (TMP) - RG 12

Swiss Regulations

Article 4 para. 4 of the Ordinance on the protection of young people in the workplace (SR 822.115) and Article 1 lit. f of the EAER regulation on hazardous work and young people (SR 822.115.2).

Take note on Article 13 Maternity Ordinance (SR 822.111.52) with regards expectant and nursing mothers.

Component	Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81)	Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC)	Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure
Dichloromethane 75-09-2 (80 - 90)	Persistent Organic Pollutants (POPs) Prohibited and Restricted Substances	Group I	

15.2. Chemical safety assessment

Chemical Safety Assessment/Reports (CSA/CSR) are not required for mixtures

Section 16: Other information

Full text of H-Statements referred to under sections 2 and 3

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H314 - Causes severe skin burns and eye damage
H318 - Causes serious eye damage
H336 - May cause drowsiness or dizziness
H351 - Suspected of causing cancer
EUH014 - Reacts violently with water
H315 - Causes skin irritation
H319 - Causes serious eye irritation

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer
Predicted No Effect Concentration (PNEC)

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (volatile organic compound)

Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:

Physical hazards On basis of test data

Health Hazards Calculation method

Environmental hazards Calculation method

Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Prepared By Health, Safety and Environmental Department

Creation Date 17-Nov-2009

Revision Date 30-Nov-2024

Revision Summary Not applicable.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006.
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No
1907/2006 .**

**For Switzerland - Compiled in accordance with the technical provisions referred to in Annex 2,
Number 3, ChemO (SR 813.11 - Ordinance on Protection against Dangerous Substances and
Preparations).**

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Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

End of Safety Data Sheet